CARBON MARKETS CCUS AND DIRECT AIR CAPTURE

CARBON-UNIT IDENTITY FORK → MARKET ARCHITECTURE MATRIX → CLAIM-TYPE LADDER → INTEGRITY GATE → CREDIT LIFECYCLE RAIL → ACCOUNTING-RISK FIREWALL

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ● SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g1 and all prior artifacts unchanged

00

STAGE 00 CARBON-UNIT IDENTITY FORK

CARBON-UNIT IDENTITY FORK · PERMISSION UNIT INSIDE A COMPLIANCE CONSTRAINT · VERIFIED OUTCOME AGAINST AN APPROVED BASELINE · ALLOCATION OR AUCTION VERSUS METHODOLOGY AND ISSUANCE · SURRENDER OR RETIREMENT UNDER THE APPLICABLE RULE · TRADABILITY DOES NOT ERASE DIFFERENT UNIT ORIGINS · MUST REMEMBER · CARBON PRICING AND MARKETS ASSIGN TRADABLE OR FISCAL INCENTIVES

ALLOWANCE → permission unit inside a compliance constraint

CREDIT → verified outcome against an approved baseline → ORIGIN → allocation or auction versus methodology and issuance → USE → surrender or retirement under the applicable rule

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• ALLOWANCE → permission unit inside a compliance constraint • CREDIT → verified outcome against an approved baseline	• ORIGIN → allocation or auction versus methodology and issuance • USE → surrender or retirement under the applicable rule	• TRAP → tradability does not erase different unit origins • MUST REMEMBER: Carbon pricing and markets assign tradable or fiscal incentives to…

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → tradability does not erase different unit origins.

01

STAGE 01 MARKET ARCHITECTURE MATRIX

MARKET ARCHITECTURE MATRIX · CAP AND TRADE · ABSOLUTE COVERED-EMISSIONS CEILING · BASELINE AND CREDIT · PERFORMANCE AGAINST TARGET OR BASELINE · LEGAL OBLIGATION AND SURRENDER REQUIREMENT · NON-MANDATED PURCHASE OR CLAIM · WHAT CONSTRAINS EMISSIONS AND WHO MUST ACT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
CAP AND TRADE → absolute covered-emissions ceiling	COMPLIANCE → legal obligation and surrender requirement	QUESTION → what constrains emissions and who must act
BASELINE AND CREDIT → performance against target or baseline	VOLUNTARY → non-mandated purchase or claim	CAP AND TRADE → absolute covered-emissions ceiling

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• CAP AND TRADE → absolute covered-emissions ceiling • BASELINE AND CREDIT → performance against target or baseline	• COMPLIANCE → legal obligation and surrender requirement • VOLUNTARY → non-mandated purchase or claim	• QUESTION → what constrains emissions and who must act • CAP AND TRADE → absolute covered-emissions ceiling

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUESTION → what constrains emissions and who must act.

02

STAGE 02 CLAIM-TYPE LADDER

CLAIM-TYPE LADDER · COUNTERFACTUAL SOURCE DOES NOT ARISE · EXISTING SOURCE EMITS LESS · CARBON LEAVES AMBIENT ATMOSPHERE · DURATION AND REVERSAL RISK DETERMINE DURABILITY · NO MERGER · EACH SUPPORTS A DIFFERENT CLIMATE CLAIM

AVOIDANCE → counterfactual source does not arise

REDUCTION → existing source emits less → REMOVAL → carbon leaves ambient atmosphere → STORAGE → duration and reversal risk determine durability

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• AVOIDANCE → counterfactual source does not arise • REDUCTION → existing source emits less	• REMOVAL → carbon leaves ambient atmosphere • STORAGE → duration and reversal risk determine durability	• NO MERGER → each supports a different climate claim • AVOIDANCE → counterfactual source does not arise

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO MERGER → each supports a different climate claim.

03

STAGE 03 INTEGRITY GATE

INTEGRITY GATE · DEFINE THE COUNTERFACTUAL · TEST BUSINESS-AS-USUAL RISK · QUANTIFY BOUNDARY PERIOD AND OUTCOME · INDEPENDENT EVIDENCE CHECK · UNIT CREATED ONLY AFTER RULE-DEFINED GATES

BASELINE → define the counterfactual → ADDITIONALITY → test business-as-usual risk → MRV → quantify boundary period and outcome → VERIFICATION → independent evidence check → ISSUANCE → unit created only after rule-defined gates

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• BASELINE → define the counterfactual • ADDITIONALITY → test business-as-usual risk	• MRV → quantify boundary period and outcome • VERIFICATION → independent evidence check	• ISSUANCE → unit created only after rule-defined gates • BASELINE → define the counterfactual

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ISSUANCE → unit created only after rule-defined gates.

04

STAGE 04 CREDIT LIFECYCLE RAIL

CREDIT LIFECYCLE RAIL · DESIGN OR VALIDATION · METHOD AND PROJECT ACCEPTED · ACTIVITY AND OUTCOME RECORDED · EVIDENCE EXAMINED · ISSUANCE AND TRANSFER · TRADABLE UNIT ENTERS REGISTRY · UNIT REMOVED FROM CIRCULATION FOR A CLAIM

DESIGN OR VALIDATION → method and project accepted → MONITORING → activity and outcome recorded → VERIFICATION → evidence examined → ISSUANCE AND TRANSFER → tradable unit enters registry → RETIREMENT → unit removed from circulation for a claim

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• DESIGN OR VALIDATION → method and project accepted • MONITORING → activity and outcome recorded	• VERIFICATION → evidence examined • ISSUANCE AND TRANSFER → tradable unit enters registry	• RETIREMENT → unit removed from circulation for a claim • DESIGN OR VALIDATION → method and project accepted

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RETIREMENT → unit removed from circulation for a claim.

05

STAGE 05 ACCOUNTING-RISK FIREWALL

ACCOUNTING-RISK FIREWALL · DOUBLE ISSUANCE · MORE THAN ONE UNIT FOR ONE OUTCOME · DOUBLE USE · ONE UNIT USED MORE THAN ONCE · DOUBLE CLAIMING · SAME OUTCOME CLAIMED BY MULTIPLE ACTORS · CORRESPONDING ADJUSTMENT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
DOUBLE ISSUANCE → more than one unit for one outcome	DOUBLE CLAIMING → same outcome claimed by multiple actors	RULE → adjustment is not issuance transfer or retirement
DOUBLE USE → one unit used more than once	CORRESPONDING ADJUSTMENT → international accounting treatment	DOUBLE ISSUANCE → more than one unit for one outcome

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• DOUBLE ISSUANCE → more than one unit for one outcome • DOUBLE USE → one unit used more than once	• DOUBLE CLAIMING → same outcome claimed by multiple actors • CORRESPONDING ADJUSTMENT → international accounting treatment	• RULE → adjustment is not issuance transfer or retirement • DOUBLE ISSUANCE → more than one unit for one outcome

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → adjustment is not issuance transfer or retirement.

06

STAGE 06 INDIA CCTS STAGE LADDER

INDIA CCTS STAGE LADDER · LEGAL BASIS · ENABLING FRAMEWORK · SCHEME DESIGN · COMPLIANCE AND OFFSET PATHWAYS · TARGET OR METHODOLOGY · OBLIGATION AND CREDIT RULES · ISSUANCE AND TRADING

LEGAL BASIS → enabling framework

SCHEME DESIGN → compliance and offset pathways → TARGET OR METHODOLOGY → obligation and credit rules → ISSUANCE AND TRADING → operational transactions

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• LEGAL BASIS → enabling framework • SCHEME DESIGN → compliance and offset pathways	• TARGET OR METHODOLOGY → obligation and credit rules • ISSUANCE AND TRADING → operational transactions	• PRICE AND VOLUME → dated market evidence only • LEGAL BASIS → enabling framework

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRICE AND VOLUME → dated market evidence only.

07

STAGE 07 CCUS CHAIN

CCUS CHAIN · CONCENTRATED CARBON DIOXIDE STREAM · CAPTURE AND CONDITIONING · SEPARATION AND PREPARATION · PIPELINE SHIP ROAD OR OTHER ROUTE · UTILISATION OR STORAGE · DESTINATION DETERMINES DURABILITY · MONITORING AND CLOSURE

SOURCE → concentrated carbon dioxide stream → CAPTURE AND CONDITIONING → separation and preparation → TRANSPORT → pipeline ship road or other route → UTILISATION OR STORAGE → destination determines durability → MONITORING AND CLOSURE → permanence and responsibility

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• SOURCE → concentrated carbon dioxide stream • CAPTURE AND CONDITIONING → separation and preparation	• TRANSPORT → pipeline ship road or other route • UTILISATION OR STORAGE → destination determines durability	• MONITORING AND CLOSURE → permanence and responsibility • SOURCE → concentrated carbon dioxide stream

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT → pipeline ship road or other route.

08

STAGE 08 UTILISATION-STORAGE TEST

UTILISATION-STORAGE TEST · PRODUCT USE · HOW LONG IS CARBON RETAINED · PROCESS USE · DOES LATER RELEASE OCCUR · GEOLOGICAL STORAGE · CONTAINMENT AND MONITORING · REVERSAL RISK AND REMEDIATION

PRODUCT USE → how long is carbon retained

PROCESS USE → does later release occur → GEOLOGICAL STORAGE → containment and monitoring → LEAKAGE → reversal risk and remediation

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• PRODUCT USE → how long is carbon retained • PROCESS USE → does later release occur	• GEOLOGICAL STORAGE → containment and monitoring • LEAKAGE → reversal risk and remediation	• NET BENEFIT → full lifecycle accounting required • PRODUCT USE → how long is carbon retained

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRODUCT USE → how long is carbon retained.

09

STAGE 09 CAPTURE TECHNOLOGY COMPARISON

CAPTURE TECHNOLOGY COMPARISON · POINT SOURCE CCUS · CARBON INTERCEPTED BEFORE RELEASE · CARBON SEPARATED FROM AMBIENT AIR · BIOGENIC CARBON PLUS CAPTURE AND STORAGE · REMOVAL TEST · ATMOSPHERIC ORIGIN PLUS DURABLE STORAGE · LIFECYCLE TEST

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
POINT SOURCE CCUS → carbon intercepted before release	BECCS → biogenic carbon plus capture and storage	LIFECYCLE TEST → energy land transport and reversal
DAC → carbon separated from ambient air	REMOVAL TEST → atmospheric origin plus durable storage	CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not…

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• POINT SOURCE CCUS → carbon intercepted before release • DAC → carbon separated from ambient air	• BECCS → biogenic carbon plus capture and storage • REMOVAL TEST → atmospheric origin plus durable storage	• LIFECYCLE TEST → energy land transport and reversal • CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not…

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Allowance is not offset, avoidance is not removal, capture is not.

10

STAGE 10 NET-ZERO MITIGATION HIERARCHY

NET-ZERO MITIGATION HIERARCHY · PREVENT UNNECESSARY EMISSIONS · DECARBONISE EXISTING ACTIVITIES · ADDRESS HARD-TO-ABATE POINT SOURCES · COUNTERBALANCE GENUINELY RESIDUAL EMISSIONS · NET BALANCE NEEDS ONE BOUNDARY AND PERIOD · EVIDENCE LIMIT · FIX SYSTEM BOUNDARY, BASELINE, UNIT

AVOID → prevent unnecessary emissions

REDUCE → decarbonise existing activities → CAPTURE → address hard-to-abate point sources → REMOVE → counterbalance genuinely residual emissions

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• AVOID → prevent unnecessary emissions • REDUCE → decarbonise existing activities	• CAPTURE → address hard-to-abate point sources • REMOVE → counterbalance genuinely residual emissions	• VERIFY → net balance needs one boundary and period • EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit…

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit,…

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Fix system boundary, baseline, unit..

11

STAGE 11 CARBON INSTRUMENTS ANSWER SPINE

CARBON INSTRUMENTS ANSWER SPINE · UNIT MARKET AND CLAIM TYPE · BASELINE MRV ISSUANCE TRANSFER RETIREMENT · CAPTURE TRANSPORT UTILISATION OR STORAGE · PERMANENCE LEAKAGE LIFECYCLE AND DOUBLE COUNTING · CURRENT RULE PRICE VOLUME COST AND CAPACITY SOURCE

CLASSIFY → unit market and claim type → TRACE → baseline MRV issuance transfer retirement → FOLLOW → capture transport utilisation or storage → TEST → permanence leakage lifecycle and double counting → AUDIT → current rule price volume cost and capacity source

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
• CLASSIFY → unit market and claim type • TRACE → baseline MRV issuance transfer retirement	• FOLLOW → capture transport utilisation or storage • TEST → permanence leakage lifecycle and double counting	• AUDIT → current rule price volume cost and capacity source • CLASSIFY → unit market and claim type

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AUDIT → current rule price volume cost and capacity source.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · DAC'S FUNDAMENTAL TECHNICAL CHALLENGE IS THERMODYNAMIC · ATMOSPHERIC CO_2 CONCENTRATION · POLICY IMPLICATION · BECAUSE DAC DELIVERS GENUINE ATMOSPHERIC REMOVAL (UNLIKE · GLOBAL CCUS AND DAC DEPLOYMENT CAPACITY REMAINS SMALL RELATIVE · THE INDIAN CARBON MARKET'S SPECIFIC SECTORAL COVERAGE, TARGET-SETTING METHODOLOGY · MRV-SYSTEM ROBUSTNESS FOR BOTH INDIA'S DOMESTIC CARBON MARKET AND

OPTIONAL SOURCE DEPTH	ADVANCED DEBATE	LEGACY / NUANCE
• CAUTION: DAC's fundamental technical challenge is thermodynamic atmospheric CO_2 concentration	• CAUTION: MRV-system robustness for both India's domestic carbon market and international	• Net-zero frameworks require genuine removal technologies (DAC/BECCS/nature-based sinks)
• CAUTION: Policy implication: because DAC delivers genuine atmospheric removal (unlike	• India's carbon market has a two-track design: a compliance mechanism (building on PAT,	• NOT: All CCUS applications, including standard fossil-power-plant CCUS, deliver negative
• CAUTION: Global CCUS and DAC deployment capacity remains small relative to the scale IPCC	• Standard fossil-source CCUS is an emissions-avoidance technology; genuine atmospheric	• NOT: India's carbon market is a single-track, purely compliance-based system. - It has a
• CAUTION: The Indian Carbon Market's specific sectoral coverage, target-setting methodology, and	• DAC's higher cost/energy-intensity relative to CCUS stems from the much lower CO_2	• NOT: DAC is currently deployed at a scale comparable to global CCUS capacity. - DAC remains

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels